

KASHIK SREDHARAN

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SUMMARY

Robotics & Embedded Systems Engineer (MSc student). Expertise in C/C++ firmware, autonomous systems, and AI integration. Proven track record in R&D, control theory, and bridging the gap between hardware and intelligent software.

WORK EXPERIENCE

Embedded Systems Engineer, Specter Products LLP.

Apr 2025 - Aug 2025

- Developed core C/C++ firmware for patented hardware, managing the full SDLC and hardware-software integration.
- Engineered low-level drivers for I2C/SPI sensors, ensuring high-speed data acquisition and synchronization.
- Executed system-level validation and unit testing to ensure 100% reliability in real-world operating conditions.

Embedded Systems Developer, Venca.

Nov 2023-Aug 2024

- Designed custom PCBs (component selection to deployment) and wrote complete embedded C firmware.
- Optimized real-time control logic, improving system response time by 20% through efficient interrupt handling.
- Implemented initial bootloaders and main application logic for a smart home automation system.

Freelance Software Developer.

Jan 2023-Sep 2023

- Implemented a REST API using JSON over HTTP to facilitate seamless data exchange between web interfaces and hardware modules.
 - Developed expertise in network protocols and data serialization, applying high-level communication standards to complex embedded system configurations.
 - Streamlined communication workflows by bridging data gaps between frontend services and backend hardware.
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EDUCATION

University of Galway, Ireland | Master of Science in Intelligent Robotics

Sep 2025- Present

- Relevant Coursework: Advanced Control Theory, Real-time Systems, Embedded Systems, IoT Integration.
- Key Project: Building an end-to-end IoT system requiring robust hardware integration and efficient data communication. The project aims for me to use an industrial grade robot to play table tennis.

Thadomal Shahani College of Engineering, Mumbai.

Dec 2021 - Mar 2025

B-Tech: Artificial Intelligence and Data Science.

PROJECTS

RoughnessAlgo: Real-Time Terrain Analysis

- Developed a novel algorithm in C++/Python for real-time terrain analysis on resource-constrained robotics hardware.
- Translated complex mathematical models into optimized firmware code, enhancing computational efficiency by 39%.

Machine Learning Model for Real-Time Stock Market Prediction

- Built a real-time predictive model using Python, scikit-learn, and TensorFlow, implementing LSTM and Random Forest architectures.
- Analyzed historical stock data to identify trends, achieving a predictive accuracy of 87%.

Table tennis playing Robot using KUKKA IIWA

- Designing a software that integrate custom kinematics for KUKKA IIWA with self written algorithm to read Neuromorphic camera data to be able to play table tennis against a human.
- The project uses custom integration software built upon Ros2, the neuromorphic camera can recognize a ball in under 1 ms and can get the speed height and spin of the ball in under 10ms giving KUKKA IIWA ample time to react to the ball.
- The KUKKA IIWA uses custom written kinematics to reach the given x,y coordinate of the plane.

ADDITIONAL INFORMATION

- **Technical Skills:** Languages: C, C++, Python, MATLAB, SQL.
- **Embedded Systems:** Microcontrollers (ARM, STM32), RTOS (concepts), Bootloaders, Firmware Development, ROS/ROS2, Gazebo, FreeRTOS.
- **Hardware Interfaces:** I2C, SPI, UART, GPIO, ADC/DAC.
- **Protocols & Data:** TCP/IP, UDP, HTTP, JSON, YAML, REST APIs.
- **Hardware Knowledge:** PCB Design (Altium/KiCad), CAD (SolidWorks), Digital Logic, Signal Processing (DSP).
- **Development Tools:** Git, GDB, Makefile, CI/CD, Linux/Unix, Simulink.
- **Soft Skills:** Communication, Collaboration, Technical Documentation, and Active Listening.